

第二周作业参考解答

p.28, 习题 10.4, 1, 2, 3, 5.

1. 解: (1) 由于 $|\sin(x^3 + y^3)| \leq |x^3 + y^3| \leq |x^3| + |y^3| \leq (|x| + |y|)(x^2 + y^2)$,

因此 $0 \leq \frac{|\sin(x^3 + y^3)|}{x^2 + y^2} \leq |x| + |y| \rightarrow 0, (x, y) \rightarrow (0, 0)$,

所以 $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0 = f(0, 0)$. 故 $f(x, y)$ 在 $(0, 0)$ 点连续.

(2) 因为 $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 1 \neq 0 = f(0, 0)$, 因此 $f(x, y)$ 在 $(0, 0)$ 点不连续.

(3) 令 $x = \rho \cos \theta, y = \rho \sin \theta$, 则当 $(x, y) \rightarrow (0, 0)$ 时, 有 $\rho \rightarrow 0$, 且

$$0 \leq |f(x, y)| = |\rho \sin^2 \theta \cos^2 \theta| < \rho \rightarrow 0,$$

故 $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0 = f(0, 0)$, 所以 $f(x, y)$ 在 $(0, 0)$ 点连续.

(4) 令 $y = x$, 则 $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$; 取 $x = y^2$, 则 $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \frac{1}{2}$, 所以

$\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ 不存在, 故 $f(x, y)$ 在 $(0, 0)$ 点不连续.

2. 解: (1) 显然, 间断点只可能出现在直线 $x + y = 0$ 上. 对该直线上的任意一点 $(x_0, -x_0)$,

若 $x_0 - x_0^2 \neq 0$, 则 $\lim_{(x,y) \rightarrow (x_0, -x_0)} f(x, y)$ 不存在, 因此函数在该点不连续; 若 $x_0 - x_0^2 = 0$, 则

$x_0 = 0$ 或 1 , 在 $(0, 0)$ 点处, 取 $y = kx$, 则 $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ 不存在, 因此函数在该点不连续;

在 $(1, -1)$ 处, 令 $u = x - 1, v = y + 1$, 则当 $(x, y) \rightarrow (1, -1)$ 时, 有 $(u, v) \rightarrow (0, 0)$, 且

$$\lim_{(x,y) \rightarrow (1,-1)} f(x, y) = \lim_{(u,v) \rightarrow (0,0)} \frac{u - v^2 + 2v}{(u + v)(v^2 + u^2 - uv + 3u - 3v + 3)},$$

当沿着直线 $u = kv$ 趋近 $(0, 0)$ 时, 上式右边极限为 $\frac{2+k}{3k+3}$ 依赖于 k . 因此 $\lim_{(x,y) \rightarrow (1,-1)} f(x, y)$

不存在, 故 $f(x, y)$ 在 $(1, -1)$ 不连续. 所以函数的间断点集是 $\{(x, y) | x + y = 0\}$.

(2) $f(x, y)$ 的间断点只可能出现在直线 $y = 0$ 上. 任取 $x_0 \neq 0$, 因为

$$\lim_{(x,y) \rightarrow (x_0,0)} f(x,y) = \lim_{(x,y) \rightarrow (x_0,0)} \frac{x_0}{y^2} e^{-\frac{x_0}{y^2}} = 0 = f(x_0,0),$$

因此 $f(x,y)$ 在 $(x_0,0)$ 点连续. 取 $x = y^2$, 则 $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} e^{-y^2} = 1 \neq 0 = f(0,0)$.

故 $f(x,y)$ 在 $(0,0)$ 不连续. 所以 $f(x,y)$ 的不连续点集为 $\{(0,0)\}$.

(3) 显然, $f(x,y)$ 在非坐标原点处连续. 下面讨论其在坐标原点处的连续性. 因为

$$0 < f(x,y) \leq \frac{e^{(x^2+y^2)^2} - 1}{x^2 + y^2},$$

且 $\lim_{(x,y) \rightarrow (0,0)} \frac{e^{(x^2+y^2)^2} - 1}{x^2 + y^2} = \lim_{(x,y) \rightarrow (0,0)} \frac{(x^2 + y^2)^2 + o((x^2 + y^2)^2)}{x^2 + y^2} = 0$, 因此

$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0 = f(0,0)$, 故 $f(x,y)$ 在 $\mathbb{R}^2$ 上连续.

3. 解: (1) 显然函数 $f(x,y)$ 的定义域是 $D = \{(x,y) | xy > -1\}$, 且 $f(x,y)$ 在 (x_0, y_0) ($x_0 \neq 0$) 点处一定连续. 故只需讨论 $f(x,y)$ 在 y 轴上的连续性即可.

(1) 在 $(0,0)$ 点, $f(0,0) = 0$.

当 $y = 0$ 时, $f(x,0) = 0$; 当 $x = 0$ 时, $f(0,y) = y \rightarrow 0$ ($(x,y) \rightarrow (0,0)$);

且 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} f(x,y) = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\ln(1+xy)}{xy} \cdot \lim_{y \rightarrow 0} y = 0$, 所以 $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0 = f(0,0)$, 故 $f(x,y)$

在 $(0,0)$ 点连续.

(2) 在 $(0, y_0)$ ($y_0 \neq 0$) 点, $f(0,y) = y \rightarrow y_0 = f(0, y_0)$ ($y \rightarrow y_0$).

$\lim_{\substack{(x,y) \rightarrow (0,y_0) \\ x \neq 0}} f(x,y) = \lim_{\substack{(x,y) \rightarrow (0,y_0) \\ x \neq 0}} \frac{\ln(1+xy)}{xy} \cdot \lim_{\substack{(x,y) \rightarrow (0,y_0) \\ x \neq 0}} y = y_0 = f(0, y_0)$;

所以 $\lim_{(x,y) \rightarrow (0,y_0)} f(x,y) = f(0, y_0)$, 从而 $f(x,y)$ 在 $(0, y_0)$ 点连续.

因此函数 $f(x,y)$ 在其定义域内连续.

(2) 显然在非坐标原点处连续. 下面讨论函数在坐标原点处的连续性.

令 $x = \rho \cos \theta$, $y = \rho \sin \theta$, 则

$$0 \leq |f(x, y)| = |\rho(\cos \theta - \sin \theta + \cos \theta \sin^2 \theta)| \leq 3\rho \rightarrow 0 \quad (x, y) \rightarrow (0, 0)$$

故 $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0 = f(0, 0)$, 因此 $f(x, y)$ 在 $(0, 0)$ 连续。

5. 证明: 因为 $\lim_{x^2+y^2 \rightarrow \infty} f(x, y) = +\infty$, 所以 $\exists \rho_1 > 0$, 使得当 $x^2 + y^2 > \rho_1$ 时, 有 $f(x, y) > f(0, 0)$. 因为 $D = \{(x, y) | x^2 + y^2 \leq \rho_1\}$ 是有界闭集, 根据有界闭集上的连续函数可以取到最小值知, 存在 $(x_0, y_0) \in D$ 使得 $f(x_0, y_0) = \min\{f(x, y) | (x, y) \in D\}$. 于是对任意 (x, y) , 若 $(x, y) \in D$, 则 $f(x, y) \geq f(x_0, y_0)$; 若 $(x, y) \notin D$, 则 $f(x, y) > f(0, 0) \geq f(x_0, y_0)$. 因此, f 在 $\mathbb{R}^2$ 上有最小值 $f(x_0, y_0)$. 证毕

p.38, 习题 11.1, 1(3), (4), (6), (8), (9), (10), 2(2), (5), (6), 3(1), (4), (6), 4, 6, 7.

$$1. \text{ 解: (3) } f'_x = \frac{xy \ln xy - (1+xy) \ln(1+xy)}{x(1+xy) \ln^2 xy}, \quad f'_y = \frac{xy \ln xy - (1+xy) \ln(1+xy)}{y(1+xy) \ln^2 xy};$$

$$(4) \quad f'_x = -\frac{2xy}{x^4 + y^2}, \quad f'_y = \frac{x^2}{x^4 + y^2};$$

$$(6) \quad f'_x = y^x \ln y + yx^{y-1}, \quad f'_y = x^y \ln x + xy^{x-1};$$

$$(8) \quad f'_x = \frac{|y|}{x^2 + y^2}, \quad f'_y = \frac{-xy}{|y|(x^2 + y^2)};$$

$$(9) \quad f'_x = \frac{z}{y} \left(\frac{x}{y}\right)^{z-1}, \quad f'_y = -\frac{zx^z}{y^{z+1}}, \quad f'_z = \left(\frac{x}{y}\right)^z \ln \frac{x}{y};$$

$$(10) \quad f'_x(x, y) = \begin{cases} \frac{2x((x^2 + y^2)y^2 \cos(x^2 y^2) - \sin(x^2 y^2))}{(x^2 + y^2)^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

$$f'_y(x, y) = \begin{cases} \frac{2y((x^2 + y^2)x^2 \cos(x^2 y^2) - \sin(x^2 y^2))}{(x^2 + y^2)^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

2. 解: (2) 利用定义, 可求得 $f'_x(0, 0) = 0 = f'_y(0, 0)$, 所以

$$\lim_{(x, y) \rightarrow (0, 0)} \frac{f(x, y) - f(0, 0) - f'_x(0, 0)x - f'_y(0, 0)y}{\sqrt{x^2 + y^2}} = \lim_{(x, y) \rightarrow (0, 0)} \frac{2xy}{x^2 + y^2},$$

取 $x = y$, 则上述极限不等于零, 故函数 $f(x, y)$ 在 $(0, 0)$ 点不可微。

(5) 由于 $\frac{\partial f}{\partial x}(0, 0) = \lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = \lim_{x \rightarrow 0} \operatorname{sgn}(x)$ 不存在;

$$\frac{\partial f}{\partial y}(0, 0) = \lim_{y \rightarrow 0} \frac{f(0, y) - f(0, 0)}{y} = \lim_{y \rightarrow 0} \frac{y^2}{y} = 0 \text{ 存在,}$$

而可微函数的偏导数一定存在, 故函数 $f(x, y)$ 在原点不可微。

(6) 容易验证,

$$\frac{|x - y|}{\sqrt{x^2 + y^2}} \leq \frac{|x| + |y|}{\sqrt{x^2 + y^2}} = \frac{\sqrt{(|x| + |y|)^2}}{\sqrt{x^2 + y^2}} \leq \sqrt{2},$$

可求得 $f'_x(0, 0) = 0 = f'_y(0, 0)$, 又 $\varphi(0, 0) = 0$, 故

$$\lim_{(x, y) \rightarrow (0, 0)} \frac{f(x, y) - f(0, 0) - f'_x(0, 0)x - f'_y(0, 0)y}{\sqrt{x^2 + y^2}} = \lim_{(x, y) \rightarrow (0, 0)} \frac{|x - y|\varphi(x, y)}{\sqrt{x^2 + y^2}} = 0,$$

所以 $f(x, y)$ 在原点可微。

$$3. \text{ 解: (1) } \left. \frac{\partial f}{\partial x} \right|_{(\frac{\sqrt{2}}{2}, \frac{1}{2}, -\frac{1}{2})} = -\frac{x}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \cos \frac{1}{\sqrt{x^2 + y^2 + z^2}} \bigg|_{(\frac{\sqrt{2}}{2}, \frac{1}{2}, -\frac{1}{2})} = -\frac{\sqrt{2}}{2} \cos 1,$$

$$\left. \frac{\partial f}{\partial y} \right|_{(\frac{\sqrt{2}}{2}, \frac{1}{2}, -\frac{1}{2})} = -\frac{y}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \cos \frac{1}{\sqrt{x^2 + y^2 + z^2}} \bigg|_{(\frac{\sqrt{2}}{2}, \frac{1}{2}, -\frac{1}{2})} = -\frac{1}{2} \cos 1,$$

$$\left. \frac{\partial f}{\partial z} \right|_{(\frac{\sqrt{2}}{2}, \frac{1}{2}, -\frac{1}{2})} = -\frac{z}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \cos \frac{1}{\sqrt{x^2 + y^2 + z^2}} \bigg|_{(\frac{\sqrt{2}}{2}, \frac{1}{2}, -\frac{1}{2})} = \frac{1}{2} \cos 1,$$

$$\text{所以 } df\left(\frac{\sqrt{2}}{2}, \frac{1}{2}, -\frac{1}{2}\right) = \left(-\frac{\sqrt{2}}{2}dx - \frac{1}{2}dy + \frac{1}{2}dz\right) \cos 1.$$

$$(4) \text{ 因为 } \frac{\partial f}{\partial x} = \frac{2x}{1 + x^2 + y^2 + z^2}, \quad \frac{\partial f}{\partial y} = \frac{2y}{1 + x^2 + y^2 + z^2}, \quad \frac{\partial f}{\partial z} = \frac{2z}{1 + x^2 + y^2 + z^2},$$

$$\text{因此 } df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz = \frac{2(x dx + y dy + z dz)}{1 + x^2 + y^2 + z^2};$$

$$(6) \text{ 因为 } \frac{\partial f}{\partial x} = -\frac{ye^{xy}}{\sqrt{1 - e^{2xy}}}, \quad \frac{\partial f}{\partial y} = -\frac{xe^{xy}}{\sqrt{1 - e^{2xy}}}, \quad \text{因此 } df = -\frac{e^{xy}(y dx + x dy)}{\sqrt{1 - e^{2xy}}}.$$

4. 解: 因为 $df(a, b) = 3dx - dy$, 因此 $f'_x(a, b) = 3$, $f'_y(a, b) = -1$, 故由函数在一点处可

微的定义知, $\lim_{x \rightarrow 0} \frac{f(a+x, b+x) - f(a, b)}{x} = \lim_{x \rightarrow 0} \frac{f'_x(a, b)x + f'_y(a, b)x + o(x)}{x} = 2$.

6. (1) 解: 圆台的体积 $V = \frac{1}{3}\pi h(R^2 + r^2 + Rr)$, 故圆台体积的增量

$$\Delta V \approx \frac{1}{3}\pi h(2R+r)\Delta R + \frac{1}{3}\pi h(2r+R)\Delta r + \frac{1}{3}\pi(R^2 + r^2 + Rr)\Delta h = 820\pi.$$

(2) 解: 令 $f(x, y, z) = \sqrt{x^2 + y^2 - z^2}$, 取 $P_0(4, 7, 1)$, 则

$$\frac{\partial f}{\partial x}(P_0) = \frac{4}{8}, \quad \frac{\partial f}{\partial y}(P_0) = \frac{7}{8}, \quad \frac{\partial f}{\partial z}(P_0) = -\frac{1}{8},$$

$$\text{故 } S \approx f(P_0) + \frac{\partial f}{\partial x}(P_0)\Delta x + \frac{\partial f}{\partial y}(P_0)\Delta y + \frac{\partial f}{\partial z}(P_0)\Delta z = 8 + \frac{0.04 + 0.21 - 0.01}{8} = 8.03.$$

7. 证明: 由于 $f(x, y)$ 关于 y 的偏导数在 $\mathbb{R}^2$ 上有界, 因此 $\exists M > 0$ 使得对 $\forall (x, y) \in \mathbb{R}^2$, 有 $|f'_y(x, y)| \leq M$. 任取 $(x_0, y_0) \in \mathbb{R}^2$. 对函数 $f(x, y)$ 关于变量 y 在以 y_0 和 y 为端点的区间上应用微分中值定理, 则存在 y' 介于 y_0 和 y 之间, 使得

$$|f(x, y) - f(x, y_0)| = |f'_y(x, y')||y - y_0| \leq M|y - y_0|.$$

任给 $\varepsilon > 0$. 因为 $f(x, y)$ 在点 (x_0, y_0) 处关于变量 x 连续, 因此 $\exists \delta > 0$ (可取 $\delta < \frac{\varepsilon}{2M}$)

使得当 $|x - x_0| < \delta$ 时, 有 $|f(x, y_0) - f(x_0, y_0)| < \frac{\varepsilon}{2}$.

对 $\forall (x, y) \in \mathbb{R}^2$, 当 $\sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$ 时, 有 $|x - x_0| < \delta$ 且 $|y - y_0| < \delta$, 从而

$$|f(x, y) - f(x_0, y_0)| \leq |f(x, y) - f(x, y_0)| + |f(x, y_0) - f(x_0, y_0)| < \varepsilon.$$

故 $f(x, y)$ 在点 (x_0, y_0) 处连续. 由点 $(x_0, y_0) \in \mathbb{R}^2$ 的任意性知 $f(x, y)$ 在 $\mathbb{R}^2$ 上连续. 证毕

p.44, 习题 11.2, 1(2), 8, 10(1), (3).

1. (2) 解: 对于 $x \neq 0$, 由于 $\frac{\partial f}{\partial y}(x, 0) = \lim_{y \rightarrow 0} \frac{f(x, y) - f(x, 0)}{y} = \lim_{y \rightarrow 0} \frac{xy}{x^2 + y^2} = 0$ 且

$$f'_y(0, 0) = \lim_{x \rightarrow 0} \frac{f(0, y) - f(0, 0)}{x} = 0, \text{ 因此 } \frac{\partial^2 f}{\partial x \partial y}(0, 0) = \lim_{y \rightarrow 0} \frac{f'_y(x, 0) - f'_y(0, 0)}{x} = 0; \text{ 但}$$

$$\frac{\partial f}{\partial x}(0, y) = \lim_{x \rightarrow 0} \frac{f(x, y) - f(0, y)}{x} = \lim_{x \rightarrow 0} \frac{y^2}{x^2 + y^2} = 1,$$

且 $f'_x(0, 0) = \lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = 0$, 从而 $\frac{\partial^2 f}{\partial y \partial x}(0, 0) = \lim_{y \rightarrow 0} \frac{f'_x(0, y) - f'_x(0, 0)}{y}$ 不存

在. 证毕

8. 证明: 当 $t \rightarrow 0$ 时, 有 $0 \leq \left| \frac{\varphi(t)}{\sqrt{t}} \right| \leq C |t|^{\mu-\frac{1}{2}} \rightarrow 0$, 故 $\lim_{t \rightarrow 0} \left| \frac{\varphi(t)}{\sqrt{2t}} \right| = 0$. 又

$$f'_x(0,0) = \lim_{x \rightarrow 0} \frac{f(x,0) - f(0,0)}{x} = 0, \quad f'_y(0,0) = \lim_{y \rightarrow 0} \frac{f(0,y) - f(0,0)}{y} = 0,$$

因此

$$\begin{aligned} 0 &\leq \lim_{(x,y) \rightarrow (0,0)} \left| \frac{f(x,y) - f(0,0) - f'_x(0,0)x - f'_y(0,0)y}{\sqrt{x^2 + y^2}} \right| \\ &= \lim_{(x,y) \rightarrow (0,0)} \left| \frac{\varphi(xy)}{\sqrt{x^2 + y^2}} \right| \leq \lim_{(x,y) \rightarrow (0,0)} \left| \frac{\varphi(xy)}{\sqrt{2|xy|}} \right| = 0, \end{aligned}$$

所以 $\lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - f(0,0) - f'_x(0,0)x - f'_y(0,0)y}{\sqrt{x^2 + y^2}} = 0$, 故 $f(x,y)$ 在 $(0,0)$ 可微。证毕

10. 解: (1) $f''_{xx} = -2a^2 \cos(2ax - 2by)$, $f''_{xy} = 2ab \cos(2ax - 2by)$,

$$f''_{yy} = -2b^2 \cos(2ax - 2by);$$

(3) $f''_{xx} = (-2y + xy^2)e^{-xy}$, $f''_{xy} = (-2x + x^2y)e^{-xy}$, $f''_{yy} = x^3e^{-xy}$.